Secured and annotated execution of workflows with WfExS-backend

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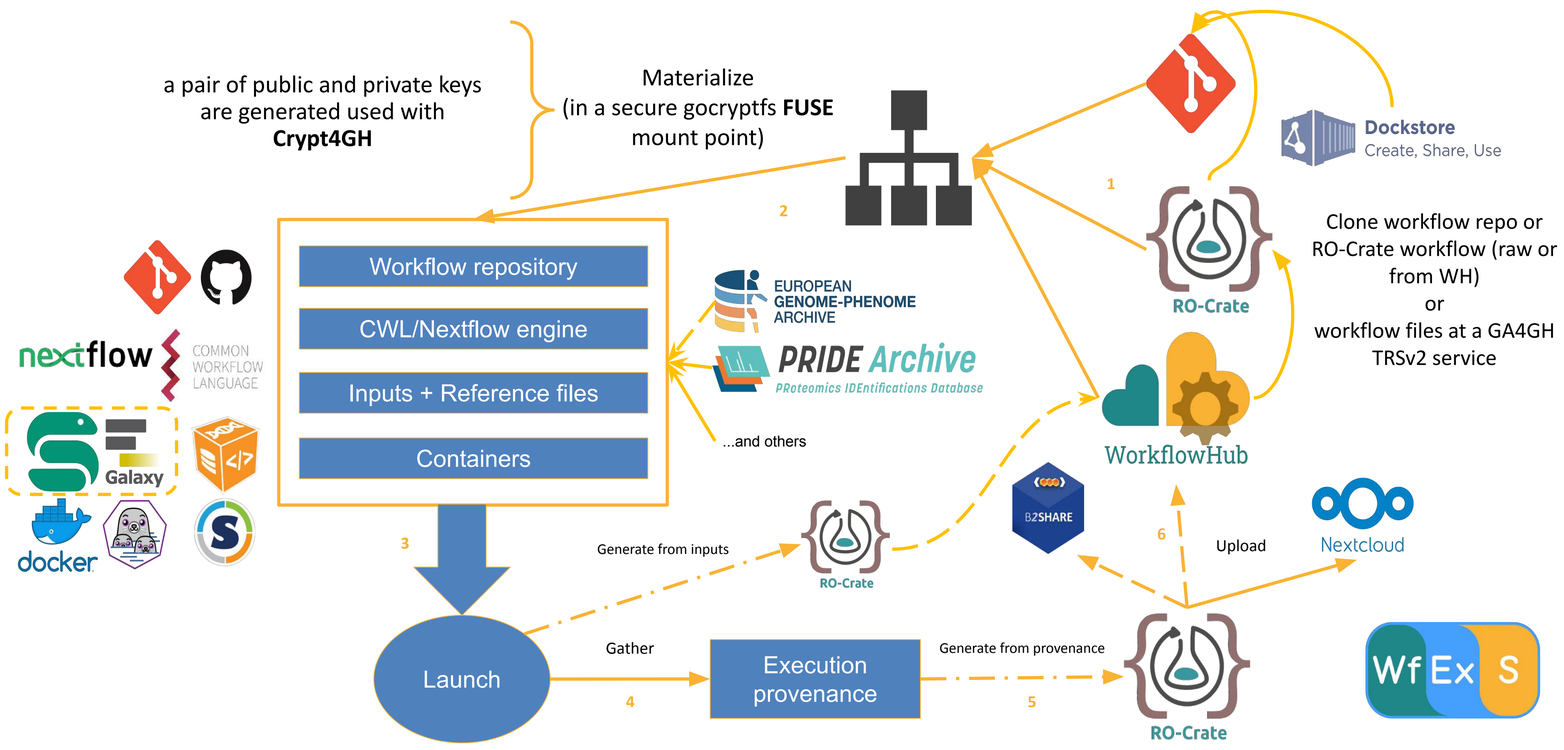
WfExS-backend (https://github.com/inab/WfExS-backend) is a high-level workflow execution orchestrator command line program, which fetches and materializes all the elements needed to run a workflow, through public or permanent identifiers: the workflow itself, the engine to run it, the needed software containers and the inputs. Workflow definitions can be fetched from either GA4GH TRS-enabled services or git repositories. As exporting results requires its own approaches, we are now writing the first pushers.

Inputs are described through URI formatted identifers which can be related to either basic, common protocols, like https, s3, gs or sftp, or to compact URIs, like encode or DRS. The orchestrator both creates and consumes RO-Crates, focusing on the interconnection of content-sensitive research infrastructures for handling sensitive human data analysis scenarios.

Analysis life-cycle using WfExS



Fetchers (schemes)
http
https
ftp
sftp
ssh
file
drs (GA4GH)
s3 (Amazon)
gs (Google)
git, git+http, git+https
pride.project
fasp* (Aspera)
trs* (GA4GH)

Pushers
cache
nextcloud

Future pushers
webdav
ftp
sftp
WorkflowHub
Zenodo
B2SHARE
osf.io

WfExS-backend delegates workflow execution on existing workflow engines, e.g. Nextflow or cwltool, and it is designed to facilitate secure and reproducible workflow executions to promote analysis reproducibility and replicability using sensitive data. Secure executions are achieved using FUSE encrypted directories for non-disclosable inputs, intermediate workflow execution results and output files.

We are extensively testing WfExS-backend in different usage scenarios and use cases. So far, next real life workflows have successfully worked using publicly available resources (reference databases, input datasets and the workflow itself). Not all the available workflows are suitable for WfExS-paradigm due reproducibility issues (i.e. their steps are not properly containerized), or dependencies on external web services.

Workflow	Topic	Kind	Project	WorkflowHub
COSIFER	RNA-seq	CWL Nextflow	Individualized PaediatricCure	118, 119
multiAffinity	RNA-seq	CWL		250
kf-alignment	Agnostic Alignment	CWL		no
Wetlab2Variations	Variant Calling RD-Connect	CWL Nextflow	EJP-RD EOSC-Life	107, 106
TCGA	Benchmarking	Nextflow	OpenEBench	244
Transproteomic	Benchmarking	Nextflow	ELIXIR Proteomics	no
MaxQuant	Benchmarking	Nextflow		no
Compomics	Benchmarking	Nextflow		no
OpenMS-ProteomicsLFQ	Benchmarking	Nextflow		no
Proline	Benchmarking	Nextflow		no
CNV_pipeline	Variant Calling	CWL	TransBioNet	278

RO-Crate representations of those executions are, indeed, an element of knowledge transfer between repeated workflow executions. WfExS-backend stores all the gathered execution details, output metadata and execution provenance in the output RO-Crate to achieve future reproducible executions. Final execution results can be encrypted with GA4GH crypt4gh standard using the public keys of the target researchers or destination so that the results can be safely moved outside the execution environments through unsecured networks and storage.

Future developments are focused on ironing secure data export procedures, supporting other workflow engines, like SnakeMake or Galaxy, and chaining of workflow executions.

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